

The background of the slide features a light blue gradient. On the left side, there is a vertical image of classical columns, possibly from a university building, rendered in a lighter blue tone. The entire slide is framed by a thin brown border.

Quantitative Synthesis of Technology

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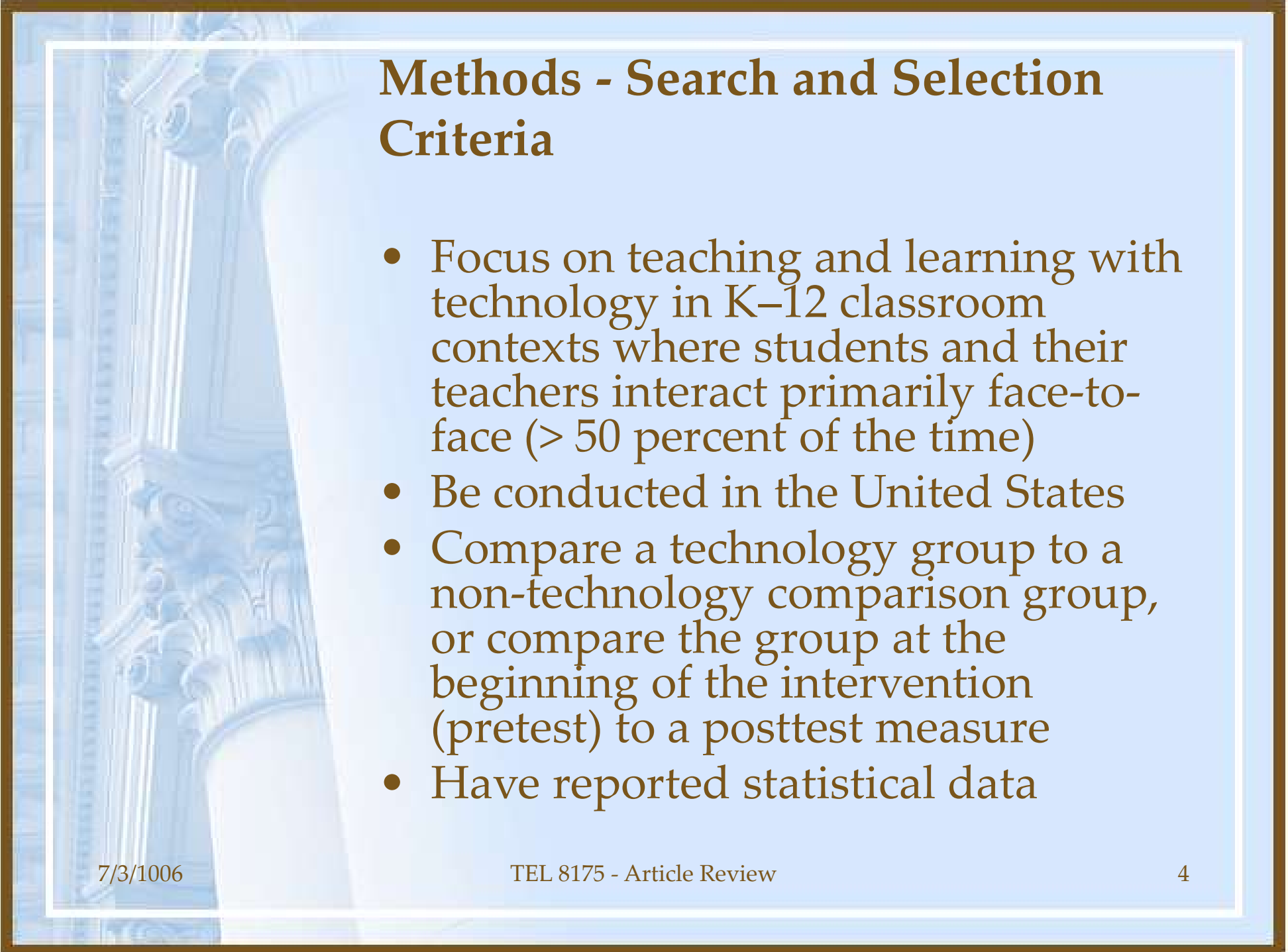
Purpose of the Study

The current study aims to synthesize recent research on the effects of technology in teaching and learning on student outcomes.

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Focus on three areas of the outcome learning

- Cognitive
- Affective
- Behavioral



Methods - Search and Selection Criteria

- Focus on teaching and learning with technology in K–12 classroom contexts where students and their teachers interact primarily face-to-face (> 50 percent of the time)
- Be conducted in the United States
- Compare a technology group to a non-technology comparison group, or compare the group at the beginning of the intervention (pretest) to a posttest measure
- Have reported statistical data

Procedure

- To calibrate the studies' results, or place them on a common scale, effect sizes were calculated.
- These effect sizes consist of the treatment group mean minus the control mean divided by the control standard deviation.
- Effect sizes can be considered a standardized estimate of where the treatment group stands in comparison with the control group distribution.

Formula

$$1/n$$

where n = number of comparisons in the study

Results

Technology

- 45% of the studies used personal computers
- 20% used networked laboratories
- 10% used multimedia
- 25% used a variety of other technology resources

Software

- 35% used an exploratory environment such as simulations, hypermedia, and hypertext
- 20% used drill-and-practice software
- 10% used tools for other tasks such as word processing or e-mail

Instructional Software

- 10% used mixed forms of technology
- 15% of the studies did not specify the software they used.

Results (cont.)

Table 2a
Summary of Mean Study-Weighted Effect Sizes for Student Outcomes

Outcomes	Number of Weighted Comparisons	Study- Weighted Effect Sizes	SD	95% Confidence Intervals	
				Lower	Higher
Cognitive	13	.390	.711	-.050	.830
Affective	5	.279	.493	-.305	.862
Behavioral	2	-.154	.402	-2.503	2.195
Overall	20	.301	.633	.004	.598

Results (cont.)

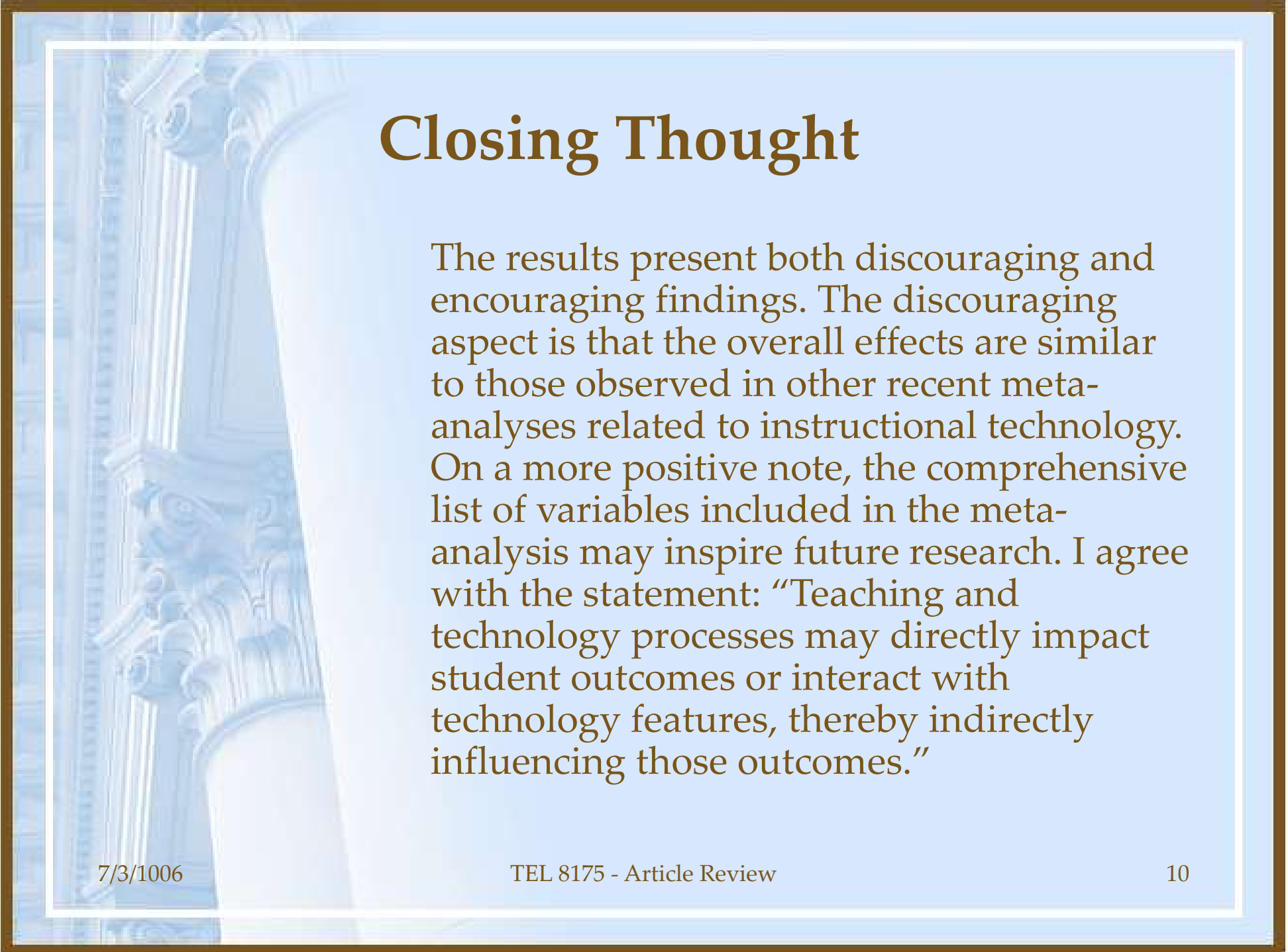
Table 2b
Summary of Unweighted Effect Sizes for Student Outcomes

Outcomes	Number of Unweighted Comparisons	Unweighted Effect Sizes	SD	95% Confidence Intervals	
				Lower	Higher
Cognitive	48	.327	.589	.156	.498
Affective	60	.208	.397	.106	.311
Behavioral	30	.096	.235	-.002	.174
Overall	138	.223	.455	.146	.229

Other recent Meta-analyses in technology

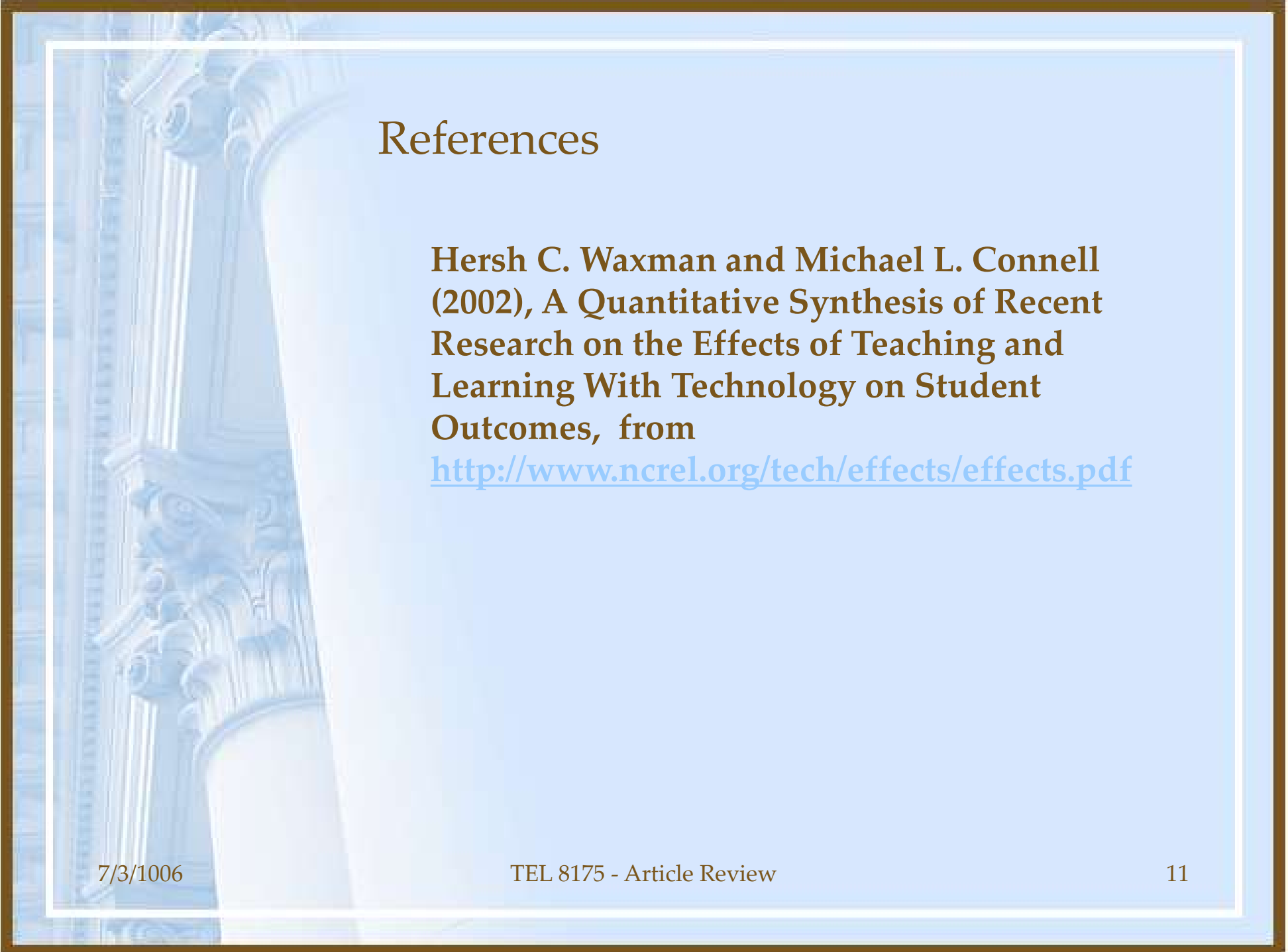
Table 1
Summary of Recent Meta-Analyses in Educational Technology

Author(s) and Date	Focus	N of Studies	Effect Size
Bayraktar (2001-2002)	CAI in secondary and college science	42	.273
Blok, Oostdam, Otter, and Overmaat (2002)	Computer-based instructional simulations	42	.190
Cavanaugh (2001)	Interactive distance education technologies	19	.147
Christmann and Badgett (1999)	CAI in science	11	.266
Christmann, Badgett, and Lucking (1997)	CAI in differing subject areas	27	.209
Christmann, Lucking, and Badgett (1997)	CAI in secondary schools	28	.172
Lee (1999)	Computer-based instructional simulation	19	.410
Lou, Abrami, and d'Apollonia (2001)	Small group v. individualized learning with technology	122	.150
Whitley (1997)	Gender differences in computer-related attitudes and behavior	82	.209
Median =		28	.209

The background of the slide features a faint, blue-tinted image of classical architectural columns, likely from a Greek or Roman temple, positioned on the left side. The columns are fluted and have ornate capitals. The overall background is a light blue gradient.

Closing Thought

The results present both discouraging and encouraging findings. The discouraging aspect is that the overall effects are similar to those observed in other recent meta-analyses related to instructional technology. On a more positive note, the comprehensive list of variables included in the meta-analysis may inspire future research. I agree with the statement: “Teaching and technology processes may directly impact student outcomes or interact with technology features, thereby indirectly influencing those outcomes.”

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References

**Hersh C. Waxman and Michael L. Connell
(2002), A Quantitative Synthesis of Recent
Research on the Effects of Teaching and
Learning With Technology on Student
Outcomes, from
<http://www.ncrel.org/tech/effects/effects.pdf>**